Laakkonen's Paradox: A Constructive Alternative to the Classical Metric of the Origin

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Language: Lean 4 Theorem Prover
GitHub: Infinitespacemechanic/Laakkonen-Paradox

Abstract — Classical field theories create singularities at geometric coordinates by conflating spatial indices with physical material states. By permitting volume and mass boundaries to drop to absolute zero (0), classical physics creates a structural paradox at the core of massive bodies where continuous downward mechanical pressure (overburden) meets an unphysical null field. This paper presents Laakkonen's Paradox and its computational resolution: a constructive framework where the ground state is structurally locked to a universal unit hold (1).

1. The Paradox of the Origin: Label vs. State

Classical $g(r)=G\cdot M(r)/r^2$ allows $M(0)=0$ at $r=0$. This treats geometric cancellation as material void. At center, qazillion pounds of overburden presses down. If g were truly 0 as state, overburden would float. r=0 is address (house number), not void. State is locked at 1.

2. Resolving the Field Metrics

Separate Net Vector Acceleration from Field Potential Intensity. Balanced vectors: $g_{net}(0)=0$ vector sum, but field intensity $g_{field}(0)=g_{max}\geq 1$, $P(0)=P_{max}>0$, $R(0)\geq 1$, $M(0)=1$. No Zero $\rightarrow$ Greater Than Zero $\rightarrow$ Holds at 1 Until Exact.

3. Radial Field Continuum

1) Origin Center r=0: $M_{hold}=1$ locked. 2) Inner Iron Zone: high-density, active grav lock $M>>1$. 3) Middle Slot: resonant containment (rings, Van Allen). 4) Gaseous Envelope: diffuse pressure accruing. 5) Dry Space: outer boundary $\lim_{r\to\infty} M(r)=1$, dry not vacuum.

4. Formal Type-Theoretic Foundations in Lean 4

LaakReal = {val: R // val≥1}. Zero is unconstructible: theorem no_zero_exists. Shifted algebras: add = a+b-1, mul = a*b, where 1 is both additive and multiplicative identity. Lean verifies add_one and mul_one.

5. Bounded Saturated Calculus

If $M(r)\rightarrow 1$ then $dM/dr\rightarrow 0$. Subtraction saturates: sub = if a-b+1≥1 then a-b+1 else 1. Theorem sub_self: a-a=1 not 0 — balance returns to rest-state. Theorem derivative_step_never_zero: (a-b).val≠0 — no differential step can be zero.

6. Conclusion

Classical treats label r=0 as value 0. LaakReal provides computer-verified non-vanishing unit space. Universe protected from null singularities at cores and boundaries; system holds at 1 until exact.